Lóránt Nagy, PhD

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Professional Experience

Eötvös Loránd University, ELTE AI Research Group **2023–Present**
Research Assistant

- Conduct research on machine-learning methods for stochastic processes and long-memory time series.
- Conducted an interpretability study of neural-network-based Hurst estimation, providing evidence that the model relies primarily on spectral rather than geometric information.
- Develop reinforcement-learning methods for trading under market frictions, comparing learned performance with theoretical asymptotic behaviour.

Alfréd Rényi Institute of Mathematics **2019–Present**
Research Fellow

- More recent work focuses on generative AI and deep learning, particularly diffusion-based generative models.
- Developed a framework for studying superlinearly mean-reverting diffusions as alternative forward processes in generative modelling, with ongoing numerical experiments comparing them with the classical dynamics.
- Conduct research in mathematical finance, stochastic control, market frictions, and optimal investment.

Artificial Intelligence National Laboratory **2024**
ML Researcher

- Worked with ALTEO on intraday electricity-price forecasting.
- Developed and evaluated machine-learning regression models and assessed their reliability.
- Communicated modelling choices, limitations, and findings to academic and industry collaborators.

Sysler Kft. **2023**
Research Consultant

- Prototyped a probabilistic late-fusion framework combining outputs from independent medical diagnostic models, based on Bayesian updating to infer disease progression from empirical patient trajectories.

Morgan Stanley **2017**
Interest Rate Modelling Intern

- Applied quantitative methods to problems in interest-rate modelling.
- Gained experience with mathematical modelling and numerical analysis in an industry setting, as well as communicating technical work within a professional team.

Education

Central European University **2018–2022**
PhD in Mathematics

Budapest University of Technology and Economics **2015–2018**
MSc in Mathematics

Technical Skills

Programming and development tools: Python, PyTorch, scikit-learn, NumPy, SciPy, Git, Docker, GitHub Actions, Wolfram Mathematica, LLM-assisted development workflows.

Quantitative methods: stochastic modelling, simulation, optimization, statistical analysis, mathematical finance, and time-series forecasting.

Cloud and deployment: Azure Static Web Apps, Azure Container Apps; basic CI/CD workflows and containerized application deployment

Machine Learning and Deep Learning tools: MLPs, RNNs; conditional VAEs, diffusion-based generative models, reinforcement-learning techniques; Weights and Biases, and TensorBoard

Selected Publications

2021 Rásonyi, M. and Nagy, L.
Optimal long-term investment in illiquid markets when prices have negative memory.
Electronic Communications in Probability, 26.

2024 Csanády, B., Nagy, L., Boros, D., Ivkovic, I., Kovács, D., Tóth-Lakits, D., Márkus, L., and Lukács, A.
Parameter estimation of long-memory stochastic processes with deep neural networks.
Proceedings of ECAI 2024.

2024 Boros, D., Csanády, B., Ivkovic, I., Nagy, L., Lukács, A., and Márkus, L.
Deep learning the Hurst parameter of linear fractional processes and assessing its reliability.
Quality and Reliability Engineering International, 40, 4228–4246.

2025 Rásonyi, M. and Nagy, L.
Optimal investment with transient price impact and stochastic spread.
Preprint, arXiv:2511.12093.

Selected Talks

2026 Bologna **XIII Bachelier World Congress**
On the utility problem in a market where price impact is transient

2025 Vienna **Vienna Congress on Mathematical Finance, TU Wien**
Non-convex optimal investment with transient price impact

2024 Szeged **Conference on Analysis and Applied Mathematics, University of Szeged**
Optimal execution in superlinearly mean-reverting environments

2023 Budapest **ELTE–Rényi AI Workshop, Eötvös Loránd University**
Risk-averse trading

2022 Ann Arbor **Financial Mathematics Seminar, University of Michigan**
Young, Timid, and Risk Takers

Additional Experience and Distinctions

2025 Intermediate-level Qualification in Copyright Law, Hungarian Intellectual Property Office.
2022 Award for Advanced Doctoral Students.
2019 Instructor, *Fundamentals of Stochastic Analysis*, Central European University.
2016 Second Place, TDK competition, *Examples in the theory of continuous trading with friction.*